# Two Programs, One Number

Race conditions and shared state

SCAN / LEARN



THE MAIN IDEA

A race condition appears when the result depends on which operation happens first. Without coordination, two valid updates can overwrite each other.

## FOLLOW THE SYSTEM

**01 START**

A shared account balance is 10.

**02 READ**

Two programs each read the value 10.

**03 CHANGE**

Each independently calculates 10 + 1.

**04 OVERWRITE**

Both write 11, even though two updates occurred.

**READ 10 → TWO UPDATES → WRITE 11 (NOT 12)**

COMMON MISCONCEPTION

Each program is correct, so their combined result must be correct.

MORE ACCURATE

Shared updates need safe ordering or atomic operations.

## THREE QUESTIONS TO KEEP

**PROBLEM:** How do simultaneous updates avoid data loss?

**FAILURE:** A change disappears without any error message.

**IN REAL LIFE:** Counters, stock levels, account balances.

SOLO DISCOVERY

Act out two readers and two writers on one number.